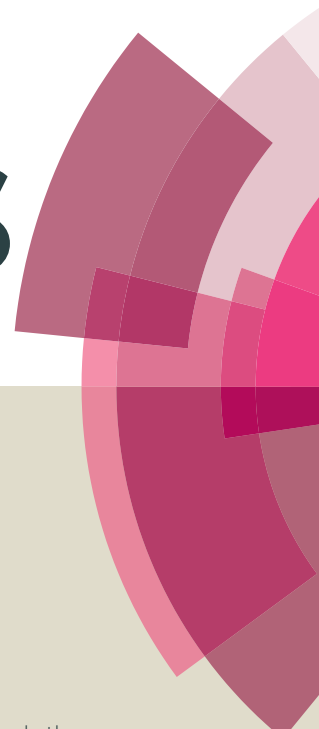


RSC Advances



This article can be cited before page numbers have been issued, to do this please use: J. Su, X. Cao, J. Wu, C. Jin, J. Tian and R. Yang, *RSC Adv.*, 2016, DOI: 10.1039/C6RA01296E.



This is an *Accepted Manuscript*, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. This *Accepted Manuscript* will be replaced by the edited, formatted and paginated article as soon as this is available.

You can find more information about *Accepted Manuscripts* in the [Information for Authors](#).

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard [Terms & Conditions](#) and the [Ethical guidelines](#) still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this *Accepted Manuscript* or any consequences arising from the use of any information it contains.

Journal Name

ARTICLE

One-pot Synthesis of Boron-doped Ordered Mesoporous Carbons as Efficient Electrocatalysts for Oxygen Reduction Reaction

 Jianmin Su^a, Xuecheng Cao^a, Jiao Wu^a, Chao Jin^a, Jing-Hua Tian^{a*} and Ruizhi Yang^{a*}

 Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

Boron-doped ordered mesoporous carbons (B-OMCs) with tunable and high level of doping content (>1wt.%) have been synthesized via an one-pot solvent evaporation induced self-assembly(EISA) process. The as-prepared B-OMCs show a highly ordered 2D hexagonal mesostructure with an average pore size of 3.4-4.0 nm, which could facilitate an efficient mass transport of O₂ and electrolyte during the oxygen reduction reaction (ORR) process. The electrochemical investigations demonstrate that B-doping could significantly enhance the electrocatalytic activity of the carbon materials for ORR in alkaline media. Specifically, the B-OMCs with boron doping content of 1.17 wt.% shows the highest electrocatalytic activity and best long-term durability for ORR as compared to the non-doped OMCs and the B-OMCs with other doping contents. Combined with various physical characterizations including X-ray diffraction, Small angle X-ray scattering, N₂ physisorption, Raman spectroscopy and X-ray photoelectron spectroscopy, the enhanced catalytic performance of the B-OMCs could be ascribed to the synergistic effects of the ordered mesostructure, specific surface area and moderate boron doping. This work not only inspires the fundamental understanding of the correlation between the catalytic performance and the morphology, structure of the OMCs and the doping extent of heteroatom, but also shows promising potential applications of the B-OMCs as efficient, low-cost catalysts in metal-air batteries and fuel cells.

1. Introduction

In order to realize the wide commercialization of fuel cells and metal-air batteries, sluggish but certainly critical catalytic process of oxygen reduction reaction (ORR) has to be overcome.^{1,2} Research efforts have been devoted to exploring highly efficient, low-cost catalysts to replace or at least partially substitute the precious metals, which has attracted abundant interest of both industry and academic community. Carbon materials doped with heteroatom, such as nitrogen (N),³⁻⁷ sulfur (S),⁸ phosphorus (P)^{9,10} and boron (B)^{11,12} have been shown as promising metal-free catalysts or catalyst supports¹³ with significantly improved catalytic performance due to the redistribution of atomic charge density and/or spin density induced by doping, which is favorable for O₂ adsorption and enhancing the ORR activity.^{14,15} It's well known that the catalytic activity of a catalyst is not only determined by the intrinsic activity of a single active site but also by the density of active sites. The density of active sites can be improved by increasing the heteroatom content doped in carbon and by designing the morphology and structure of the carbon materials.¹⁶ Carbon materials with larger specific surface area, optimized pore size distribution and morphology are proven to be important for good catalytic performance.¹⁷ Hence, the highly ordered

mesoporous carbons (OMCs) have received considerable attention due to their tunable pore size and highly ordered pore structure with super large specific surface area.¹⁸⁻²⁰ The ordered mesoporous structure with high specific surface area can not only provide high exposure of active sites to reactants but also facilitate an efficient mass transport of O₂ and electrolyte during the ORR process. Bo et al.²¹ studied the ORR catalytic activity of boron-doped OMCs prepared by using co-impregnation and carbonization approach with sucrose, 4-hydroxyphenylboronic acid as the carbon, boron sources and SBA-15 silica as hard template. The fabricated B-OMCs exhibit excellent catalytic activity for ORR in alkaline medium with higher selectivity and better long-term stability than the commercial Pt/C catalysts. Whereas, the traditional hard template approach usually involves multiple synthesis steps, which is time-consuming and not suitable for large-scale production. In contrast, the soft-templating method is more feasible and scalable.^{22,23} Wickramaratne and Jaroniec²² reported the synthesis of boron-doped OMCs by soft-templating method. However, no catalytic performance of this material for ORR was provided. Pašti et al.²³ prepared OMCs doped with low levels (<1 at.%) of B, N and P via soft-templating method. The ORR activity measurements reveal that each doped OMCs displays higher activity compared to non-doped one, and the activity increases in the order N-OMCs < P-OMCs < B-OMCs. Nevertheless, the doping level in their work is quite low (<1 at.%) and the stability of the doped OMCs hasn't been investigated yet.

Since the catalytic activity of the heteroatom-doped carbon materials is closely related to the morphology, structure, specific surface area of the carbon and the doping level of the

^a College of Physics, Optoelectronics and Energy & Collaborative Innovation Center of Suzhou Nano Science and Technology, Soochow University, Suzhou 215006, China. Email: jhtian@suda.edu.cn; yangrz@suda.edu.cn. Tel.: +86 512 65227675, Fax: +86 512 65221519.

Electronic Supplementary Information (ESI) available: See DOI: 10.1039/x0xx00000x

heteroatoms. Herein, we report the synthesis of boron-doped OMCs with tunable, high level of doping content (>1 wt.%) via a soft-templating method and investigate the ORR activity and stability of the B-OMCs in alkaline media thoroughly. The electrochemical studies reveal that the B-OMCs with boron doping content of 1.17 wt.% shows the highest electrocatalytic activity and best long-term durability for ORR, compared to the non-doped OMCs and B-OMCs with other boron doping contents. Combined with other physical characterizations, the results demonstrate that the ordered mesostructure, specific surface area and the doping content of boron play crucial roles and work synergistically on modifying the ORR catalytic pathway and enhancing the ORR activity of boron-doped OMCs.

2. Experimental

2.1 Synthesis of B-OMCs

The B-OMCs were synthesised through a slightly modified recipe of evaporation induced self-assembly (EISA) method developed by Zhao's group,^{20,24} using phenolic resin and H_3BO_3 as the carbon and boron sources, respectively. The triblock copolymer F127 (poly(ethyleneoxide)-b-poly(propylene oxide)-b-poly(ethylene oxide) (PEO-PPO-PEO) was used as a soft template and ethanol as a solvent. Scheme 1 depicts the synthesis procedure schematically. Typically, 1.22 g of phenol was melted at 40 °C, and then 0.26 g of NaOH solution (20 wt.%) was added slowly to the melted phenol under stirring. Then 2.1 g of formalin (37 wt.%) was added dropwise to the above mixed solution at 70 °C with stirring for 60 min. After cooling to room temperature, the pH of the mixture was adjusted to neutral by using 0.6 M HCl solution. After drying, the precursor was re-dissolved in ethanol, mixed with 2.0 g of F127 and different content of H_3BO_3 (0.2–1.0 g). Transparent composite films were obtained after transferring the solution to Petri dishes and evaporating the ethanol at 40 °C for 4 h. Then the films were dried in an oven at 100 °C for 24 h to thermo-polymerize the phenolic resins followed by calcination at 900 °C for 4 h under N_2 atmosphere. Finally, the samples were washed with hot water to get rid of the soluble boron oxide. By simply changing

the mass ratio of phenolic resin, H_3BO_3 and F127, the mesostructure, specific surface area and boron doping content in B-OMCs can be tuned facilely. The obtained samples were denoted as B-C-x (x = 1–5), where x from 1 to 5 represents the dosage of H_3BO_3 with 0.2, 0.4, 0.6, 0.8 and 1.0 g, respectively. Pure mesoporous carbon (denoted as C) without boron doping was also prepared with the same procedure without adding of H_3BO_3 for comparison.

2.2 Physical Characterizations

X-ray diffraction (XRD) was used to examine the crystal structure of the samples by a Bede D1 X-ray diffractometer (UK, Bede Scientific Ltd; $Cu K\alpha$ radiation; operated at 40 kV, 45 mA; $\lambda = 0.15418$ nm), the diffraction angle ranging from 10° to 80° with a step of 0.02° and a rate of 1.2° min⁻¹.

Small angle X-ray scattering (SAXS) measurements were taken on a Nanostar U small-angle X-ray scattering system (Bruker, Germany) using $Cu K\alpha$ radiation (40 kV, 35 mA). The d-spacing values were calculated from the formula: $d = 2\pi/q$. The unit-cell parameters were calculated from the formula $a = 2d_{10}/\sqrt{3}$.²⁵

The specific surface area of the samples is determined by nitrogen sorption isotherm, which was performed on an Automated Area and Pore Size Analyzers (Quantachrome, QuadraSorb SI).²⁶ The pore size distribution was evaluated with BJH method.²⁷ Before measurements, the samples were degassed at 200 °C for 4 h under vacuum.

The morphology and microstructure of the samples were examined by scanning electron microscopy (SEM, Hitachi S4800) and transmission electron microscopy (TEM, TecnaiG220, operating at 200 kV), respectively.

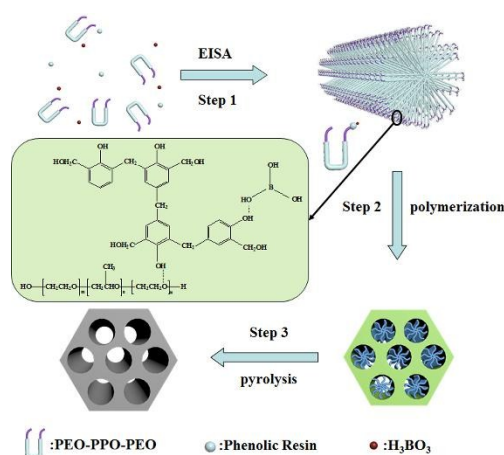
The doping content of boron in OMCs was determined by inductively coupled plasma-atomic emission spectroscopy analysis (ICP-AES, Vista MPX).

Raman spectra of the samples were examined by a Jobin Yvon LabRAM HR 800 instrument with a 514 nm excitation laser at a power of around 1 mW.

Surface analysis of the products was characterized by a SSI (Surface Science Instruments) X-ray photoelectron spectroscopy (XPS) spectrometer equipped with a hemispherical analyzer and using a monochromatized Al $K\alpha$ (1486 eV) source with a 250×1000 μm illumination spot.

2.3 Electrochemical measurements

The catalytic performance of the catalysts was performed with rotating ring-disk electrode (RRDE) technique on an electrochemical system (AFMSRX rotator, and AFCBP1bipotentiostat). The RRDE electrode is made of glass carbon disk (diameter of 5 mm with geometric surface area of 0.196 cm²), surrounded by a Pt ring (geometric surface area of 0.125 cm²). Electrochemical tests by using a traditional three-electrode system were performed at room temperature. The working electrode was equipped with a glass carbon electrode covered with the slurry of catalyst. Typically, the catalyst slurry was made by mixing 10 mg catalyst powder, 95 μL Nafion solution (5 wt.%) and 350 μL ethanol, then dispersed by



Scheme 1 Schematic illustration of the synthesis of B-OMCs.

ultrasonication for 40 min. Finally, 7 μL of the catalyst ink was dropped onto the glassy carbon disk by pipette, which results in a powder loading of $1006 \mu\text{gcm}^{-2}$. A Pt-wire and a standard Ag/AgCl electrode (3 M Cl⁻, Cypress) were used as the counter electrode and the reference electrode, respectively. The electrolyte was 0.1 M KOH solution prepared from ultrapure water (Millipore, $18.2 \text{ M}\Omega\text{cm}$). Before electrochemical measurements for each catalyst, the solution was degassed with high purity of nitrogen for more than 30 min. Electrochemical cleaning of the catalyst on GC disk was operated in N₂-saturated 0.1 M KOH solution by potential sweeping in the range of -0.8 - 0 V (vs. Ag/AgCl) at 50 mVs^{-1} , until getting stable cyclic voltammograms (CV). Finally, the CV curves were measured in the potential range of -0.8 - 0 V with scanning speed of 10 mVs^{-1} , which could be used to determine the non-Faradaic ($i_{\text{N}_2\text{-CV}}$) current density. Before the voltammetric tests of RRDE for the ORR, the electrolyte was degassed by purging high purity of oxygen for more than 30 min to make the solution saturated with O₂. Linear sweep voltammetry test of ORR was performed in O₂-saturated 0.1 M KOH by sweeping potential in the range of -0.8 - 0 V at 10 mVs^{-1} under a rotating speed of 400, 900, 1600 and 2500 rpm, respectively. Faradaic current density can be obtained by the following formula:

$$j = -(j_{\text{ORR}} - j_{\text{N}_2\text{-CV}})/SA_{\text{geo}} \quad (1)$$

Where j_{ORR} is current density of the ORR, SA_{geo} is geometric surface area of GC disk (0.196 cm^2). According to Koutecky-Levich equation, the kinetic current density of ORR could be derived from the following formula:

$$1/j = 1/j_k + 1/j_d = 1/j_k + 1/(B\omega^{1/2}) \quad (2)$$

Where j is the disk current density, j_k and j_d are the kinetic and diffusion limiting current densities, respectively. B represents the Levich slope, which is stemmed from the following equation:

$$B = 0.62nFD_{\text{O}_2}^{2/3} \nu^{-1/6} C_{\text{O}_2} \quad (3)$$

Where n is the transferred electron number in the ORR process, F represents the Faraday constant ($F = 96,485 \text{ Cmol}^{-1}$), D_{O_2} is the diffusion coefficient of oxygen in solution ($D_{\text{O}_2} = 1.86 \times 10^{-5} \text{ cm}^2\text{s}^{-1}$), ν is the kinetic viscosity of electrolyte solution ($\nu = 0.01 \text{ cm}^2\text{s}^{-1}$), C_{O_2} is the concentration of O₂ dissolved in electrolyte ($C_{\text{O}_2} = 1.21 \times 10^{-6} \text{ molcm}^{-3}$).^{28,29} The ring potential was held at 0.5 V (vs. Ag/AgCl) to oxidize the intermediate product of HO₂⁻ in alkaline solution during the process of oxygen reduction reaction.³⁰ The produced amount of %HO₂⁻ and the electron number n transferred during ORR could be calculated with the following equations³¹⁻³³:

$$\% \text{HO}_2^- = 100 \frac{2j_{\text{R}}/N}{\text{abs}(j_{\text{D}}) + (j_{\text{R}}/N)} \quad (4)$$

$$n = 4 \frac{\text{abs}(j_{\text{D}})}{\text{abs}(j_{\text{D}}) + (j_{\text{R}}/N)} \quad (5)$$

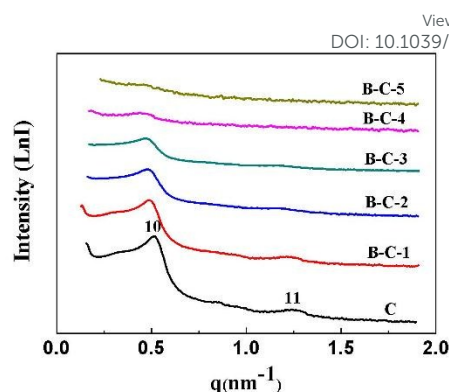


Fig. 1 SAXS patterns of the pure mesoporous carbon (C) and the mesoporous carbons doped with different content of boron (B-C-1, B-C-2, B-C-3, B-C-4 and B-C-5).

Where j_{D} is the Faradic current density at GC disk, j_{R} is the Faradic current density at Pt-ring, and N represents the collection efficiency ($N = 0.22$). The onset potential, at which the current for oxygen reduction is first observed, was determined by the point of intersection of two tangent lines, one drawn parallel to the baseline (i.e. from approx. -0.1V to 0 V vs. Ag/AgCl) and the second parallel to the increasing current signal.

3. Results and Discussion

By varying the doping content of boron in the mesoporous carbon, a series of B-OMCs were synthesized. Figure 1 shows the SAXS patterns of the pure mesoporous carbon (C) and the mesoporous carbons doped with different content of boron (B-C-1, B-C-2, B-C-3, B-C-4, B-C-5). From the bottom line (black line) in Fig. 1, we can see that the SAXS pattern of the pure mesoporous carbon exhibits two resolved scattering peaks located at 0.51 nm^{-1} and 1.25 nm^{-1} , which are corresponding to the 10 and 11 planes of the OMCs, respectively. The SAXS pattern of the C indicates that the pure mesoporous carbon has a highly ordered 2D hexagonal (p6m) mesostructure, which is well consistent with the previous report.²⁰ The intense peak of 10 plane reflects a d-spacing of 12.3 nm, which corresponds to the cell parameter of $a_0 = 14.2 \text{ nm}$. Similar to the non-doped OMCs, B-C-1, B-C-2 and B-C-3 with different boron doping contents also show resolved SAXS patterns. However, the intensity of the peaks are gradually weakened and the peak positions shift to the lower q with the increasing of doping content of boron, suggesting that the mesostructures are slightly disordered and the cell parameters are increased. When the doping content of boron was further increased (B-C-4, B-C-5), the peaks of SAXS are hardly to be distinguished, which implies that the highly ordered hexagonal mesostructure has been destroyed with the excess H_3BO_3 during the assembling process. This phenomenon has also been observed and well documented in other literatures for boron doped OMCs.^{22,24}

SEM and TEM were performed to determine the effect of boron doping on the morphology and porosity of OMCs. Here, B-C-1 and B-C-5 are taken as examples, which are shown in Figure 2. SEM images (Fig. 2a and 2b) and TEM images (Fig. 2c

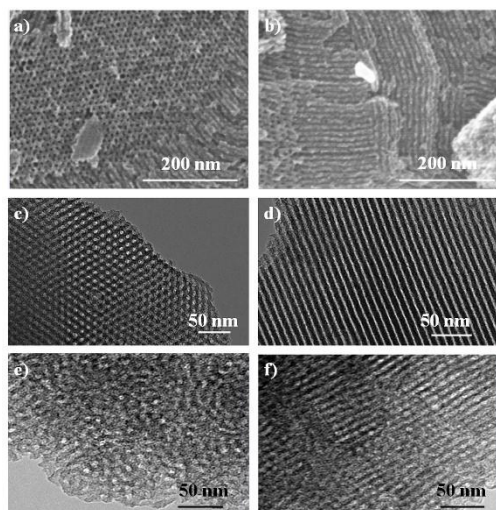


Fig. 2 SEM images (a, b) and TEM images (c, d) of the B-C-1. TEM images (e, f) of the B-C-5.

and 2d) of the sample B-C-1 exhibit well-ordered hexagonal arrays of mesopores structure and 1D channels with a pore width of approximately 3.4 nm, which confirms the formation of a 2D ordered hexagonal mesostructure. The sample B-C-5 made by using the highest doping amount of H_3BO_3 exhibits an obviously deteriorated structure with disordered mesopores, shown as the TEM image in Fig. 2e). The cross section view of the B-C-5 (Fig. 2f) further confirms that the 1D channels with a pore width of about 4.0 nm are obviously distorted, which agree with the SAXS results well (Fig. 1) and indicates that the doping with boron could affect the morphology and the porosity of the OMCs significantly

XRD profiles of the B-OMCs are shown in Fig. 3a). The pure OMCs displays two relatively sharp diffraction peaks located at about 24° and 43° , which are corresponding to planes of (002) and (100), respectively. All the B-OMCs show broader and weaker diffraction peaks compared to that of the pure carbon at the same positions, indicating the gradually distorted mesostructure of the ordered mesoporous carbon with the increasing amount of B-doping, which is in line with the SEM,

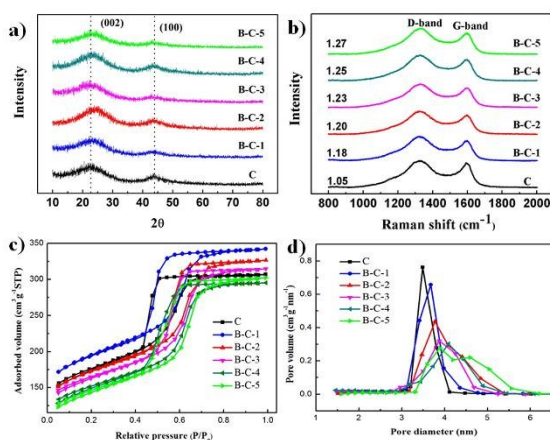


Fig. 3 a) XRD patterns, b) Raman spectra, c) BET surface areas and d) pore size distributions of the C, B-C-1, B-C-2, B-C-3, B-C-4 and B-C-5.

Table 1 Textural Properties and B doping contents of C and B-C-X (X = 1, 2, 3, 4, 5). DOI: 10.1039/C6RA01296E

Sample	BET surface area / m^2g^{-1}	Total micropore volume / cm^3g^{-1}	Average pore size / nm	ICP B% (weight)
C	565	0.47	3.36	--
B-C-1	626 ↑	0.53 ↑	3.39 ↓	0.75 ↓
B-C-2	558 ↓	0.51 ↓	3.62 ↓	1.17 ↓
B-C-3	526 ↓	0.49 ↓	3.70 ↓	1.26 ↓
B-C-4	485 ↓	0.46 ↓	3.77 ↓	1.56 ↓
B-C-5	470 ↓	0.45 ↓	3.98 ↓	1.77 ↓

TEM and SAXS results. In order to further characterize the structure defect degree of the B-OMCs, the Raman spectroscopy measurements were performed on different samples. The comparison of Raman spectra between carbon and B-doped carbons are exhibited in Fig. 3b). The intensity ratio between D-band and G-band (I_D/I_G) is used as a descriptor for the defect degree of the carbon materials, which are located approximately at 1330 cm^{-1} and 1600 cm^{-1} , respectively.²³ As the boron doping content increases, the I_D/I_G of the samples increases gradually. The value of I_D/I_G is 1.05 for the sample C and it increases from 1.18 to 1.27 for the B-OMCs, which indicates that more defects are introduced in the OMCs after being doped with more boron, consisting with the results of SAXS, SEM, TEM and XRD. The nitrogen adsorption-desorption isotherms and the pore size distributions of all the samples are shown in Fig. 3c) and 3d), respectively. The isotherms of all the samples exhibit typical type-IV curve with a pronounced capillary condensation step, characteristic of mesoporous materials. And the pore size distribution curves indicate only mesopores exist in the synthesized B-OMCs. The ordered pure mesoporous carbon shows a BET surface area of $565\text{ m}^2\text{g}^{-1}$ and a pore volume of $0.47\text{ cm}^3\text{g}^{-1}$. The average pore size is centered at 3.36 nm as can be seen from the pore size distribution. The surface area of the B-C-1 increases to $626\text{ m}^2\text{g}^{-1}$ and the pore volume of it is $0.53\text{ cm}^3\text{g}^{-1}$. This could be ascribed to the doping effect of H_3BO_3 precursor, which is known as a modifier agent.¹⁹ The average pore size of the B-C-1 is centered at 3.39 nm. With the further increasing doping amount of H_3BO_3 , the BET surface area decreases from 558 to $470\text{ m}^2\text{g}^{-1}$, the pore volume decreases from 0.51 to $0.45\text{ cm}^3\text{g}^{-1}$, and the pore size of the samples expands gradually from 3.62 to 3.98 nm (from B-C-2 to B-C-5). The textural parameters of all the samples are listed in Table 1. The trends could be easily followed with the red dotted arrows aside.

The precisely doping contents of boron in various B-OMCs determined from the ICP-AES analysis are 0.75, 1.17, 1.26, 1.56, 1.77 wt% as the amount of H_3BO_3 increases from 0.2 to 1.0 g, respectively (Table 1), which indicates that boron has been doped into the carbon materials successfully and plays significant role on the microstructure, specific surface area, pore size and pore volume on the carbon materials.

XPS was used to investigate the surface chemical composition and the boron species doped in B-OMCs. Here, B-C-2 is taken as an example, the results are shown in Figure 4. The XPS survey spectrum of B-C-2 displays that the sample contains B, C, O and In elements, which are shown in Fig. 4a). The peak at binding energy of 192 eV corresponds to B1s. The

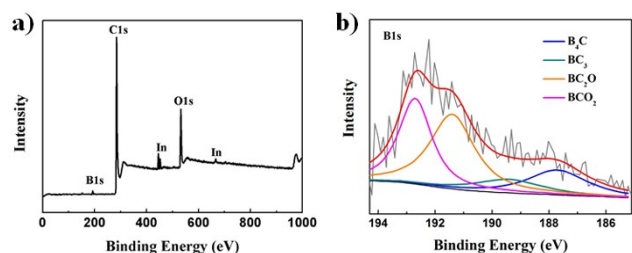


Fig. 4 a) XPS spectrum of the B-C-2, b) High-resolution B1s core-level XPS spectrum of the B-C-2.

small peaks at around 445 eV and 665 eV belong to the metal indium (In), which was used as the substrate for XPS measurements. As shown in Fig. 4b), the B1s spectrum of the sample B-C-2 could be deconvoluted into four components: the peak at the highest binding energy of 192.7 eV is attributed to BCO_2 , the peak at 191.4 eV is assigned to BC_2O , and the peaks centered at 189.5 eV and 187.7 eV correspond to the BC_3 and B_4C , respectively.^{19,34} According to the previous reports,¹³ the four type boron bonds in the B-OMCs have different contributions to electrocatalytic activity. The B_4C and B_3C types would play a key role on the improvement of the electrochemical activity of B-OMCs. Firstly, the graphite-like B_4C changes the valence band structure and improves the density of states close to the Fermi level of carbon-based materials, which could enhance the electronic conductivity of the carbon material.³⁵ Secondly, the introduction of boron in carbon lattice, such as B_4C or BC_3 doped types, increases the number of hole-type charge carriers and the electron density of nearby active carbon sites, which could also improve the conductivity and electrochemical performance of the B-doped carbons.³⁶ The contents of BCO_2 and BC_2O doped types are relatively high because boron prefer to bond with oxygen atoms than carbon atoms.³⁷ The high-resolution B1s core-level XPS spectra of the other boron-doped carbons (B-C-1, B-C-3, B-C-4, B-C-5) shown the similar results, which could be found in the Fig. S1 in the supplementary information.

The electrocatalytic performances for ORR of C and B-C-x are investigated with RRDE technique and the results are shown in Figure 5. The ORR activity of the commercial Pt/C (20%) is also included for better comparison.³⁸ From Fig. 5a), we could see that the diffusion limiting current density of the pure mesoporous carbon material (C) could reach -5.17 mAcm^{-2} and the onset potential is about -0.23 V . With the doping of boron, the catalytic activity of the sample B-C-x is remarkably improved, characterized with the positive shifting of the onset potentials for ORR and higher diffusion limiting current densities. Among the various doping contents, the sample B-C-2 exhibits the best ORR activity, characterized with the highest diffusion limiting current (-5.42 mAcm^{-2}) and the lowest onset potential (-0.14 V), which is 90 mV positively shifting compared to that of the non-doped OMCs. The achieved diffusion limiting current here is much higher than that of the previously reported B-OMCs.^{21,23} The half-wave potential of B-C-2 is about 120 mV negatively shifted compared to that of the commercial Pt/C,

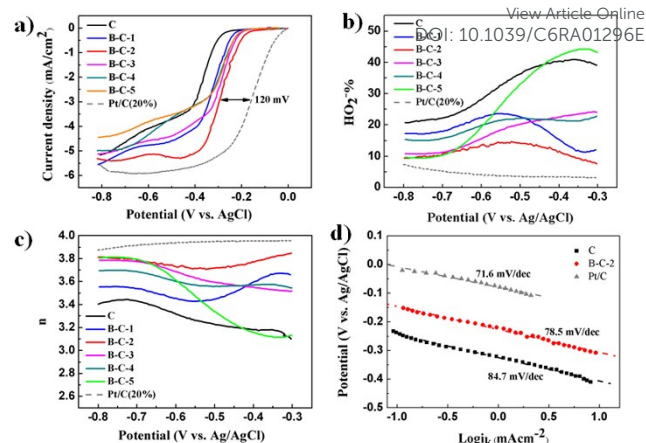


Fig. 5 a) LSVs of ORR on C, B-C-1, B-C-2, B-C-3, B-C-4 and B-C-5, b) generated HO_2^- and c) transferred electron number n calculated based on RRDE data, d) Tafel plots of C, B-C-2 and Pt/C derived from the data in fig. a).

which is in good agreement with the values reported elsewhere^{21,39} and the activity of B-C-2 electrocatalyst compared to other known B-doped carbons could be found in Table S1 in the supplementary information. In order to further understand the reaction kinetics during the ORR process, the generated peroxide species (HO_2^-) and the transferred electron number (n) during the reaction are recorded and analyzed, which are shown in Fig. 5b) and 5c), respectively. From Fig. 5b), we could see that the generated amount of HO_2^- is 21–41% for the non-doped OMCs in the potential range of -0.8 – -0.3 V , and decreases significantly after being doped with boron. The transferred electron number of the B-OMCs is also increased obviously. It is worth noting that the generated HO_2^- is less than 14% and the transferred electron number is about 3.71–3.85 for B-C-2, which is close to that of the commercial Pt/C. The diffusion-current-corrected Tafel curves of the C, B-C-2 and Pt/C are illustrated in Fig. 5d), as calculated based on the corresponding RRDE data in Fig. 5a). The kinetic current densities were calculated from the Tafel plots using equation (2). At low over-potentials, the sample C shows a slope of 84.7 mVdec^{-1} . The sample B-C-2 exhibits a smaller Tafel slope of 78.5 mVdec^{-1} and approaches that of the commercial Pt/C (71.6 mVdec^{-1}), which is close to the theoretical value of $2.303RT/F$ (*i.e.*, 59 mVdec^{-1} at 25°C). Here, R is the universal gas constant, T is the absolute temperature and F is the Faraday constant. The highest intrinsic catalytic activity of sample B-C-2 is further verified from its lowest Tafel slope.^{40–42} All the results indicate that the oxygen reduction pathway for the OMCs has been modified with the B-doping, and the electrocatalytic activity for ORR has also been promoted significantly.

The polarization curves for the ORR on B-C-2 at different rotating rates are exhibited in Fig. 6a). All the curves reach the well-defined diffusion limiting current density. A small reduction peak at about -0.3 V under a rotating rate of 400 rpm most likely results from the reduction process of the adsorbed oxygen molecules in the mesopores of B-C-2.^{17,43,44} Fig. 6b) displays the corresponding Koutecky-Levich plots measured from the inverse current density (j^{-1}) as a linear relationship to the inverse of the square root of the rotating rate ($\omega^{-1/2}$) at -0.35 ,

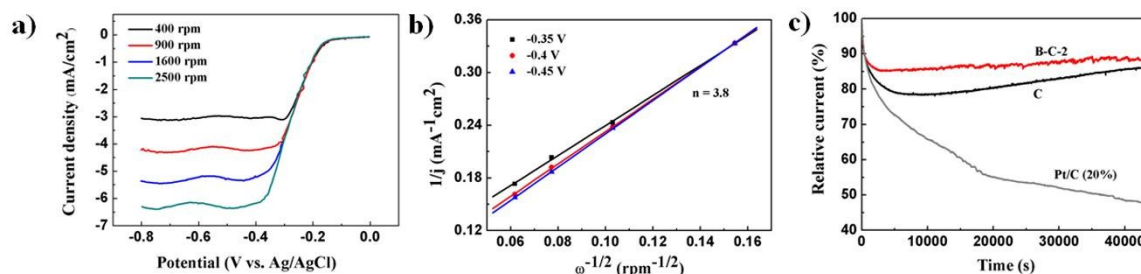


Fig. 6 a) Rotating-disk cyclic voltammograms of the B-C-2 under rotation speed of 400, 900, 1600 and 2500 rpm, respectively, b) The corresponding Koutecky-Levich plots at different potentials, c) The chronoamperometric responses of C, B-C-2 and the commercial Pt/C kept at -0.3 V in O_2 -saturated 0.1 M KOH solution.

-0.40 and -0.45 V, respectively. These plots are nearly parallel, which confirms the first-order dependence of the kinetics of ORR process on the surface of B-C-2. The transferred electron number calculated from the K-L slope is approximately 3.8. This is completely consistent with the result ($n = 3.71$ -3.85) calculated from the RRDE measurements, indicating that the ORR process on the surface of B-C-2 is mainly a quasi-four electron pathway. The stabilities of the samples for the ORR process were also investigated by the chronoamperometric method as exhibited in Fig. 6c). And that of the commercial Pt/C is also included for better comparison. The ORR current densities of C and B-C-2 are declined generally by 22% and 15% over 12 h with continuous operation, respectively. Whereas, the commercial Pt/C suffers a serious attenuation of 53% due to the migration and agglomeration of Pt nanoparticles during the long-term reaction. And we also found that the ORR current density of the pure carbon material reduces first and then increases unexpectedly, which is probably ascribed to the

oxidation of carbon during the ORR process in alkaline solution.³⁶ However, the B-C-2 shows obviously superior stability to that of the C. The significant oxidation resistance of the boron-doped carbon may result from the strong covalent bonding between C and B,⁴⁰ which has been deduced from the XPS results (Fig. 4b).

Along with the excellent ORR activity, the B-C-2 also shows superior stability and tolerance towards methanol crossover, which is shown as Fig. 7. The chronoamperometric responses to methanol introduced into the O_2 -saturated electrolyte were performed for B-C-2 and Pt/C catalysts at -0.25 V. After the addition of 1.5 mL of methanol at 1200 s, the ORR relative current for B-C-2 shows a slightly decrease ($\sim 2.5\%$). However, the ORR relative current for the commercial Pt/C catalyst suffers a sharp decrease ($\sim 70\%$). Since the B-C-2 in our work does not possess the highest surface area, neither the highest doping content of boron (Table 1), which implies that an optimized ordered mesostructure and suitable boron doping content could affect the electrocatalytic activity of OMCs synergistically. This is probably due to the fact that higher content of boron could cause the distortion of mesoporous structure of carbon and induce more defect sites, which may destroy the ordered hexagonal structure and reduce the electrical conductivity of B-OMCs.^{45,46}

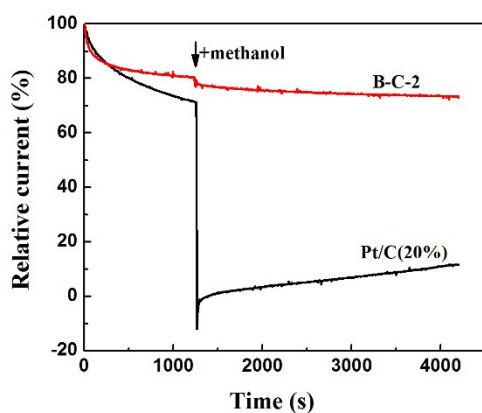


Fig. 7 The chronoamperometric responses to methanol introduced into the O_2 -saturated electrolyte for B-C-2 and Pt/C catalysts.

4. Conclusions

In summary, we report a facile and controllable approach to prepare ordered mesoporous carbon doped with tunable and high level of boron (B-OMCs) in one-pot via a solvent evaporation induced self-assembly process. The obtained B-OMCs possess tunable mesostructure and pore size, high specific surface area (470 - $626\text{ m}^2\text{g}^{-1}$) and moderate boron doping content (0.75 - $1.77\text{ wt.}\%$), which could be achieved just by simply adjusting the ratio of phenolic resol, F127 and H_3BO_3 .

The electrochemical investigations demonstrate that B-doping could significantly enhance the electrocatalytic activity of the carbon materials for ORR in alkaline media. Specifically, the sample B-C-2 doped with 1.17 wt.% boron exhibits the best ORR activity, characterized with the lowest onset potential (-0.14 V), highest diffusion limiting current (-5.42 mAcm⁻²) and smallest Tafel slope (78.5 mVdec⁻¹). The long-term stability of the B-C-2 for ORR is also superior to that of both the non-doped carbon and the commercial Pt/C. Besides, the B-C-2 exhibits excellent stability and tolerance towards methanol crossover. All the results reveal that the order degree of the mesostructure, specific surface area and the doping content of boron play important roles and work synergistically on modifying the ORR catalytic pathway and enhancing the ORR activity of the mesoporous carbons, which inspire the fundamental understanding of the correlation between the catalytic performance and the morphology, structure of the OMCs and the doping level of heteroatom. It is believed that the B-doped ordered mesoporous carbons are promising candidates as versatile nanomaterials for multiple applications, such as fuel cells and metal-air batteries.

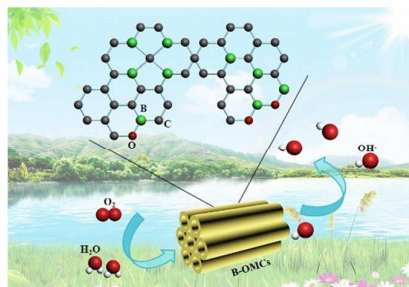
Acknowledgements

This work was supported by National Natural Science Foundation of China (Nos.51272167, 21206101, 51572181 and 21303114) and Natural Science Foundation of the Higher Education Institutions of Jiangsu Province, China (15KJB480001).

References

- M.S. Dresselhaus and I.L. Thomas, *Nature*, 2001, **414**, 332.
- P. G. Bruce, L. J. Hardwick and K. M. Abraham, *Mater. Res. Soc. Bull.*, 2011, **36**, 506.
- K. P. Gong, F. Du, Z. H. Xia, M. Durstock and L. M. Dai, *Science*, 2009, **323**, 760.
- R. Liu, D. Wu, X. Feng and K. Mullen, *Angew. Chem. Int. Ed.*, 2010, **49**, 2565.
- G. Wu, K. L. More, C. M. Johnston and P. Zelenay, *Science*, 2011, **332**, 443.
- L. P. Zhang and Z. H. Xia, *J. Phys. Chem. C*, 2011, **115**, 11170.
- B. Y. Xia, Y. Yan, N. Li, H. B. Wu, X. W. Lou and X. Wang, *Nat. Energy*, 2016, **1**, 15006.
- Z. Yang, Z. Yao, G. Li, G. Fang, H. Nie, Z. Liu, X. Zhou, X. Chen and S. Huang, *ACS Nano*, 2012, **6**, 205.
- Z. W. Liu, F. Peng, H. J. Wang, H. Yu, J. Tan and L. L. Zhu, *Catal. Commun.*, 2011, **16**, 35.
- R. Li, Z. D. Wei, X. L. Gou and W. Xu, *RSC Adv.*, 2013, **3**, 9978.
- L. Yang, S. Jiang, Y. Zhao, L. Zhu, S. Chen, X. Wang, Q. Wu, J. Ma, Y. Ma and Z. Hu, *Angew. Chem. Int. Ed.*, 2011, **50**, 7132.
- G. Jo and S. Shanmugam, *Electrochem. Comm.*, 2012, **25**, 101.
- H. Wang, L. Thia, N. Li, X. Ge, Z. Liu and X. Wang, *ACS Catal.*, 2015, **5**, 3174-3180.
- M. Zhang and L. M. Dai, *Nano Energy*, 2012, **1**, 514.
- J. P. Paraknowitsch and A. Thomas, *Energy. Environ. Sci.*, 2013, **6**, 2839.
- W. Yang, T. P. Fellingner and M. Antonietti, *J. Am. Chem. Soc.*, 2011, **133**, 206.
- P. H. Matter, L. Zhang and U. S. Ozkan, *J. Catal.*, 2006, **239**, 83.
- Y. Meng, D. Gu, F. Q. Zhang, Y. F. Shi, L. Cheng, D. Feng, Z. X. Wu, Z. X. Chen, Y. Wan, A. Stein and D. Y. Zhao, *Chem. Mater.*, 2006, **18**, 4447.
- D.-W. Wang, F. Li, Z.-G. Chen, G. Q. Lu and H.-M. Cheng, *Chem. Mater.*, 2008, **20**, 7195. DOI: 10.1039/C6RA01296E
- Y. Meng, D. Gu, F. Q. Zhang, Y. F. Shi, H. F. Yang, Z. Li, C. Z. Yu, B. Tu and D. Y. Zhao, *Angew. Chem. Int. Ed.*, 2005, **44**, 7053.
- X. J. Bo and L. P. Guo, *Phys. Chem. Chem. Phys.*, 2013, **15**, 2459.
- N. Wickramaratne and M. Jaroniec, *RSC Adv.*, 2012, **2**, 1877.
- I. Pašti, N. Gavrilov, A. Dobrota, M. Momčilović, M. Stojmenović, A. Topalov, D. Stanković, B. Babić, G. Čirić-Marjanović and S. Mentus, *Electrocatal.*, 2015, **6**, 498.
- J. Wei, D. D. Zhou, Z. K. Sun, Y. H. Deng, Y. Y. Xia and D. Y. Zhao, *Adv. Funct. Mater.*, 2013, **23**, 2322.
- F. Q. Zhang, Y. Meng, D. Gu, Y. Yan, Z. X. Chen, B. Tu and D. Y. Zhao, *Chem. Mater.*, 2006, **18**, 5279.
- J. J. Duan, S. Chen, S. Dai and S. Z. Qiao, *Adv. Funct. Mater.*, 2014, **24**, 2072.
- E. P. Barrett, L. G. Joyner and P. P. Halenda, *J. Am. Chem. Soc.*, 1951, **73**, 373.
- N. M. Markovic, H. A. Gasteiger and P. N. Ross, *J. Phys. Chem.*, 1996, **100**, 6715.
- D. R. Lide, CRC handbook of chemistry and physics, CRC Press, Boca Raton, 1995.
- J. Sunarso, A. A. Torriero, W. Zhou, P. C. Howlett and M. Forsyth, *J. Phys. Chem. C*, 2012, **116**, 5827.
- Y. Y. Liang, H. L. Wang, J. G. Zhou, Y. G. Li, J. Wang, T. Regier and H. J. Dai, *J. Am. Chem. Soc.*, 2012, **134**, 3517.
- J. Du, Y. D. Pan, T. R. Zhang, X. P. Han, F. Y. Cheng and J. Chen, *J. Mater. Chem.*, 2012, **22**, 15812.
- J. Wu, Z. R. Yang, X. W. Li, Q. J. Sun, C. Jin, P. Strasser and R. Z. Yang, *J. Mater. Chem. A*, 2013, **1**, 9889.
- K. Ghosh, M. Kumar, T. Maruyama and Y. Ando, *Carbon*, 2009, **7**, 1565.
- Q. H. Yang, W. H. Xu, A. Tomita and T. Kyotani, *J. Am. Chem. Soc.*, 2005, **127**, 8956.
- H. Shi, Y. F. Shen, F. He, Y. Li, A. Liu, S. Q. Liu and Y. J. Zhang, *J. Mater. Chem. A*, 2014, **2**, 15704.
- C. Wang, Z. Y. Guo, W. Shen, Q. J. Xu, H. M. Liu and Y. G. Wang, *Adv. Funct. Mater.*, 2014, **24**, 5511.
- C. H. Huang, Q. Zhang, T. C. Chou, C. Chen, D. S. Su and R. A. Doong, *ChemSusChem*, 2012, **5**, 563.
- L. Jiang, A. Hsu, D. Chu and R. Chena, *J. Electrochem. Soc.*, 2009, **156**, B643.
- F. Y. Cheng and J. Chen, *Chem. Soc. Rev.*, 2012, **41**, 2172.
- N. Leonard, V. Nallathambi and S. C. Barton, *J. Electrochem. Soc.*, 2013, **160**, F788.
- J. P. Paraknowitsch, Y. J. Zhang, B. Wienert and A. Thomas, *Chem. Commun.*, 2013, **49**, 1208.
- J. H. Jiang and B. L. Yi, *J. Electroanal. Chem.*, 2005, **577**, 107.
- J. Wu, Z. R. Yang, Z. J. Wang, Q. J. Sun, R. Z. Yang, *Electrochem. Commun.*, 2014, **42**, 46.
- Z. R. Ismagilov, A. E. Shalagina, O. Y. Podyacheva, A. V. Ischenko, L. S. Kibis, A. I. Boronin, Y. A. Chesalov, D. I. Kochubey, A. I. Romanenko, O. B. Anikeeva, T. I. Burvakov and E. N. Tachev, *Carbon*, 2009, **47**, 1922.
- H. Niwa, M. Kobayashi, K. Horiba, Y. Harada, M. Oshima, K. Terakura, T. Ikeda, Y. Koshigoe, J. Ozaki, S. Miyata, S. Ueda, Y. Yamashita, H. Yoshikawa and K. Kobayashi, *J. Power Sources*, 2011, **196**, 1006.

Graphical Abstract



Boron-doped OMCs with a high doping content of 1.17 wt.% shows excellent catalytic performance for ORR in alkaline media.